InnCad Plus — Quick Start

Version 0.10.0 · free parametric CAD for 3D printing, on the FreeCAD 1.1.3 engine

> The program's menus and buttons are in English; this guide gives their names as you see them on the screen.

1 Install

macOS (Apple Silicon, macOS 13 or newer)

1. Open the DMG and drag InnCadPlus.app onto Applications.
2. First launch: macOS cannot verify the app. Click **Done** (not Move to Trash).
3. System Settings ▸ Privacy & Security ▸ **Open Anyway**, confirm with your password. Needed only once.

Windows 10 / 11 (x64) — Portable

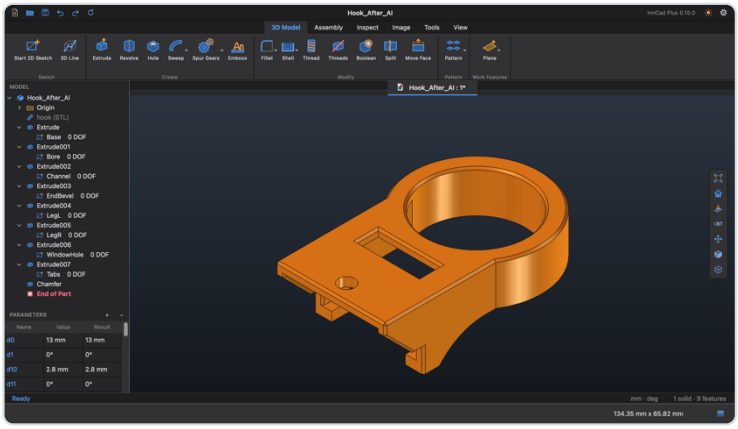
1. No installation needed: unzip the ZIP to a short path with Latin letters only, for example `C:\InnCad`.
2. Run **InnCadPlus.exe**.
3. "Windows protected your PC"? Click **More info ▸ Run anyway**.

No FreeCAD to install: the engine is inside. Downloads: github.com/Igr11/InnCadPlus-releases/releases/latest

2 First start

InnCad asks what to make: **Part** (one part) or **Assembly** (part files placed and constrained), and the units: **Millimeters** or **Inches**. Click **Create**. With **Remember my choice** it does not ask again.

3 The window



The ribbon on top (tabs **3D Model**, **Inspect**, **Tools**, **View**); the **Model Browser** on the left with the part's history down to **End of Part**; **Parameters** below it — every dimension by name, formulas allowed; the status line at the bottom: tips in blue, errors in red.

4 Mouse

Left click	Select; ⇧ or ⌘ (Ctrl on Windows) adds to the selection
Left + right drag	Orbit around the point under the cursor
Middle drag	Pan
Wheel	Zoom at the cursor
Right click	Marking menu; a quick right drag runs its command in that direction
Double click	In the browser: edit that feature or sketch; in a sketch: edit a dimension

No middle button? **Orbit** and **Pan** on the strip at the right edge of the view.

5 Your first part in five steps

1. Press S (**Start 2D Sketch**) and click **XY · Top**.
2. Press R (**Rectangle**) and click two corners.
3. Press D (**Dimension**), click a side, type **40** and press Enter; the other side **20**.
4. **Finish Sketch** (or Esc).
5. Press E (**Extrude**), set **Distance** to **10 mm**, **OK**. Then **Fillet** on a few edges.

Every step is a row in the **Model Browser**: double-click it to change it. Undo is ⌘Z (Ctrl+Z on Windows).

6 Try next

A photo at its true size

Image ▸ Insert Image, then **Scale**: two points and one real length. **Horizontal** / **Vertical** levels a crooked photo.

Real threads

3D Model ▸ Modify ▸ Thread on a round face. **Print-friendly** has flat crests; **Print clearance** of any size (0.2 mm by default) lets printed parts screw together. **Custom**: your own pitch, angle and depth — for jars and lids.

An STL back into a part

Tools ▸ Files ▸ Import the STL, then **Mesh Section** or **Smooth** — or let your AI rebuild it.

Your AI in InnCad

Install Claude Code, Codex or Gemini CLI, then **Tools ▸ AI ▸ Connect AI**. Tell it what you want, in your words.

3D lines

3D Model ▸ Sketch ▸ 3D Line: pick points on a virtual cube, on any layer — a path for **Sweep** and **Loft**.

Too big for the printer

Draw the cut line (a dovetail, for example) in a sketch, **Split ▸ Keep all**, then **Make Assembly**: the pieces become an assembly at once, each a part to print. A second line with **Offset** gives any clearance (delete the strip between).

7 Print it

File ▸ Export: **STL** for your slicer (the **STL Precision** slider sets how fine) or **STEP** for other CAD programs. The files stay plain FreeCAD .FCStd.

8 Help

- The full manual (English, Russian) explains every tool; chapter 10 has tips for 3D printing.
- Something wrong? ⚙ ▸ **Show Log File** shows the log; send it with what you did: github.com/Igr11/InCad/issues

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InnCad Plus is made by one person. Wishes, bug reports and requests are welcome — please forgive me in advance if fixing a bug or making a wish come true takes a while.